

Arnav Mahadev

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EDUCATION

University of California, Berkeley

Bachelor of Science in Electrical Engineering and Computer Sciences - GPA: 3.7/4.0

Berkeley, CA

Expected May 2028

RELEVANT COURSEWORK

Machine Learning, Data Structures, Efficient Algorithms, Operating Systems, Database Systems, Computer Architecture, Internet Architecture, Discrete Math, Probability Theory, Linear Algebra, Multivariable Calculus, Circuits & Devices

EXPERIENCE

iPick.ai

Aug 2026 – Present

Machine Learning Engineering Intern

Berkeley, CA

- Built an LLM-based portfolio optimizer using **Claude API** that segments holdings into investment tracks, prunes dominated positions, and provides per-stock and per-portfolio risk metrics
- Engineered a **reinforcement learning pipeline** of three trading agents (power picks, hidden gems, and a short-target detector) that uses realized returns over a fixed holding window as the reward signal to update each agent's policy

Software Engineering Intern

Jan 2026 – May 2026

- Architected a graph-based company relationship visualizer using **React, D3.js, and PostgreSQL**
- Extracted supplier and subsidiary relationships across **5,000+ public company tickers**, storing their structured edges in **PostgreSQL** to power graph-based visualization in the frontend
- Integrated the **Claude API** to generate real-time news summaries, enriching company profiles with live information

Maadhev LLC

May 2026 – Aug 2026

Software Engineering Intern

Cupertino, CA

- Built a custom ACH payment system (Python, Flask, Stripe API) integrated with a Shopify app, enabling bank transfer payments at **70% lower per-transaction cost** than card processing
- Automated order confirmation via **Stripe webhook handlers** that sync payment and fulfillment data between Stripe and Shopify in real time, eliminating manual reconciliation across every order

Claythis

May 2025 – Aug 2025

Software Engineering / Game Development Intern

San Mateo, CA

- Created a **Unity-based** game in **C#** showcasing Claythis's **AI-driven 3D model generation pipeline**, implementing core weapon & health systems such as reloading, recoil, animations, armor, and healing
- Validated integration of the **AI-generated 3D character models**, verifying rigging, animation, and rendering behavior

PROJECTS

Falsetto | *Python, PyTorch, Hugging Face Transformers, scikit-learn, Gradio, Modal, librosa*

Jul 2026 – Sep 2026

- Developed an AI-generated music detector using MERT embeddings and an attention-based AudioCAT classifier
- Achieved **87.5% accuracy and 0.972 AUC** with an end-to-end ML pipeline, validated by **56 automated tests**

SoccerBoard | *Python, JavaScript, XGBoost, PyTorch, FastAPI, Docker*

May 2026 – Jul 2026

- Trained a match predictor backtested on **2,500+ matches**, improving outcome accuracy from **47.5% to 59.9%**
- Developed an expected goals model matching StatsBomb's professional benchmark **within 3.2%** of actual goals

SaveBites | *JavaScript, Python, React, HTML/CSS*

Sep 2025 – Dec 2025

- Built the grocery-tracking backend for a food-waste app that scans receipts to auto-populate ingredient inventory
- Developed a **RESTful API** for full CRUD on **MongoDB** ingredient data, fed by an **OCR pipeline** for receipts

LEADERSHIP

Vice President of Professional Development | *Open Project Berkeley*

Aug 2025 – Present

- Ran resume workshops, mock interviews, and **50+ 1:1 resume reviews** for a club of **300 technical consultants**

Soccer Captain | *De Anza Force 06B NPL & Monta Vista Varsity Soccer*

Aug 2021 – Jun 2024

- Captained my school and club teams, scored **40+ goals** over three seasons as **3x All-League** and **league top scorer**

Eagle Scout | *Scouts BSA Troop 476*

Mar 2017 – Feb 2024

- Led a classroom painting project, managing **15 volunteers**, mentored **5 scouts** and completed **200 service hours**

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C#, SQL

Backend: Node.js, Express, Flask, FastAPI, REST

Frontend: React, HTML/CSS, Liquid

ML/Data: XGBoost, PyTorch, scikit-learn, NumPy, pandas

Data/Cloud: PostgreSQL, MongoDB, AWS, Docker

Tools: Git, pytest, JUnit